



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
YEAR 11
YEARLY EXAM

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 60

- Attempt questions 1-3
- Hand up in 3 sections clearly marked A,B & C

Examiner: *A.M.Gainford*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SECTION A

Question 1 (20 marks)

Marks

- a) Find the remainder when the polynomial $P(x) = x^3 + 3x - 2$ is divided by $x - 3$. **1**
- b) If $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$, find the possible values of $\tan \theta$. **2**
- c) The equation $x^2 - (1 - 2k)x + k + 3 = 0$ has consecutive integral roots. Find the possible values of k . **3**
- d) i) In how many ways can 6 committee members be selected from 10 people? **4**
- ii) In how many ways can this be done if two particular people will only serve together?
- e) Express $\sin 3\theta$ as an expression in powers of $\sin \theta$ only. **2**
- f) i) Write the equation of the tangent to the curve $y = x^3$ at the point on the curve where $x = 1$. **3**
- ii) Find the co-ordinates of the point where this tangent crosses the curve.
- g) Solve $|2x + 6| < 4$ **2**
- h) Solve for x : $\left(x + \frac{1}{x}\right)^2 - \left(x + \frac{1}{x}\right) - 12 = 0$ **3**

SECTION B**Question 2 (20 marks)****Marks****Start a New Booklet**

- a) Find the coordinates of the point which divides AB with $A(1,4)$ and $B(5,2)$ externally in the ratio 1:3. **2**
- b) Given the curve with equation $y = \frac{x}{x^2 + 1}$: **7**
- i) Find the first and second derivatives.
 - ii) Identify and determine the nature of any turning points and points of inflexion.
 - iii) Make a neat sketch of the curve.
- 3**
- c) Solve the inequality $\frac{4}{5-x} \geq 1$. **4**
- d) Find the general solution to the equation $\cos(\theta + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$. **4**
- e)
- i) Express $\sin \theta + \sqrt{3} \cos \theta$ in the form $R \sin(\theta + \alpha)$.
 - ii) Hence or otherwise sketch the graph of $y = \sin x + \sqrt{3} \cos x$ in the domain $0 \leq x \leq \pi$.

SECTION C

Question 3 (20 marks)

Marks

Start a New Booklet

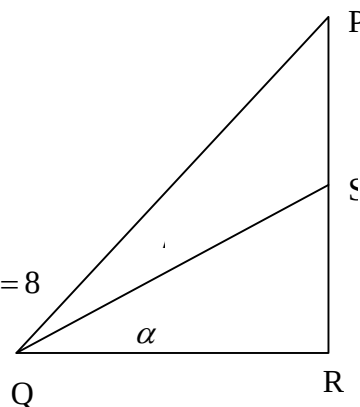
- a) In the diagram $\angle \alpha > \angle \beta > 0$ and $\frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta)} = \frac{4}{5}$

8

i) Show that $\tan \alpha \cdot \tan \beta = \frac{1}{9}$

ii) If $PR = QR$, show that $9(\tan \alpha + \tan \beta) = 8$

iii) Hence find α and β .



4

- b) i) The polynomial equation $P(x) = 0$ has a double root at $x = 9$. By writing $P(x) = (x - a)^2 Q(x)$, where $Q(x)$ is a polynomial, show $P'(a) = 0$.

- ii) Hence or otherwise find the values of a and b if $x = 1$ is a double root of $x^4 + ax^3 + bx^2 - 5x + 1 = 0$

- c) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

8

- i) Derive the equation of the tangent to the parabola at P .
- ii) Find the coordinates of the point of intersection T of the tangents to the parabola at P and Q .
- iii) Given that the tangents at P and Q in (ii) intersect at an angle of 45° , show that $p - q = 1 + pq$.
- iv) By Evaluating the expression $x^2 = 4ay$ at T , or otherwise, find the locus of T .

SECTION A. (Q1).

$$(a) R = P(3) = 27 + 9 - 2 \\ = \boxed{34} \quad \boxed{1}$$

$$(b) \operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$$

$$\therefore \sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ, 120^\circ \text{ etc.}$$

$$\therefore \boxed{\tan \theta = \pm \sqrt{3}} \quad \boxed{2}$$

(c) Let the roots be α & $\alpha+1$.

$$\text{Now } \alpha + (\alpha+1) = -\frac{b}{a} = 1-2k \quad \text{ie } 2\alpha+1 = 1-2k \\ \text{ie } \alpha = -k.$$

$$\& \alpha(\alpha+1) = \frac{c}{a} = k+3 \quad \text{ie. } -k(-k+1) = k+3 \\ k^2 - k = k+3$$

$$k^2 - 2k - 3 = 0$$

$$(k+1)(k-3) = 0$$

$$\boxed{k = 3, -1} \quad \boxed{3}$$

$$(d) (i) \binom{10}{6} = \boxed{210} \quad \boxed{2}$$

(ii) If they (ie 2 particular people) will only serve together, they are either both on or both off.

$$\therefore \binom{8}{4} + \binom{8}{6} = 70 + 28 = \boxed{98} \quad \boxed{2}$$

$$\begin{aligned}
 (a) \quad \sin 3\theta &= \sin(2\theta + \theta) \\
 &= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\
 &= 2\sin \theta \cos \theta \cdot \cos \theta + (1 - 2\sin^2 \theta) \sin \theta \\
 &= 2\sin \theta \cos^2 \theta + \sin \theta - 2\sin^3 \theta \\
 &= 2\sin \theta (1 - \sin^2 \theta) + \sin \theta - 2\sin^3 \theta \\
 &= 2\sin \theta - 2\sin^3 \theta + \sin \theta - 2\sin^3 \theta \\
 \therefore \sin 3\theta &= 3\sin \theta - 4\sin^3 \theta
 \end{aligned}$$

2

$$\begin{aligned}
 (f) \quad (i) \quad y &= x^3 \\
 y' &= 3x^2 \quad \therefore \text{slope of tangent at } x=1, \\
 &\quad \text{is } 3.
 \end{aligned}$$

$\therefore$ eqn. of tangent at $(1, 1)$

$$\text{is } y - 1 = 3(x - 1)$$

$$\begin{aligned}
 y - 1 &= 3x - 3 \\
 \underline{y} &= \underline{3x - 2}
 \end{aligned}$$

$1\frac{1}{2}$

$$(ii) \quad \text{Solving } y = x^3 \text{ and } y = 3x - 2.$$

$$\text{ie } x^3 = 3x - 2$$

$$x^3 - 3x + 2 = 0 \quad \text{--- (A)}$$

now (A) will have roots $1, 1$ & α

$$\text{now } 1 + 1 + \alpha = -\frac{b}{a} = 0.$$

$$\therefore \alpha = -2.$$

$\therefore$ the other point is $(-2, -8)$

$1\frac{1}{2}$

$$(g) \quad |2x+6| < 4$$

$$\therefore -4 < 2x+6 < 4$$

$$-10 < 2x < -2$$

$$\underline{-5 < x < -1}$$

2

$$(h) \quad \text{let } x + \frac{1}{x} = u.$$

$\therefore$ equation becomes

$$u^2 - u - 12 = 0.$$

$$(u-4)(u+3) = 0$$

$$u = 4, -3$$

$$\text{If } u = 4$$

$$x + \frac{1}{x} = 4$$

$$x^2 + 1 = 4x$$

$$x^2 - 4x + 1 = 0.$$

$$x = \frac{4 \pm \sqrt{16-4}}{2}$$

$$= \frac{4 \pm \sqrt{12}}{2}$$

$$= \frac{4 \pm 2\sqrt{3}}{2}$$

$$\underline{x = 2 \pm \sqrt{3}}$$

$$\text{If } u = -3$$

$$x + \frac{1}{x} = -3.$$

$$x^2 + 1 = -3x.$$

$$x^2 + 3x + 1 = 0.$$

$$x = \frac{-3 \pm \sqrt{9-4}}{2}$$

$$\underline{x = \frac{-3 \pm \sqrt{5}}{2}}$$

3

YR 11 Sept 2006 ext 1 paper

Section B

Q 2. (a) $A(\overset{x_1}{1}, \overset{y_1}{4})$ $B(\overset{x_2}{5}, \overset{y_2}{2})$ m n
 $-1:3$

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$\Rightarrow \left(\frac{-1 \times 5 + 3 \times 1}{-1 + 3}, \frac{-1 \times 2 + 3 \times 4}{-1 + 3} \right) = (-1, 5) \quad (2)$$

(b) $y = \frac{x}{x^2+1}$

(i) $y' = \frac{(x^2+1) \times 1 - x \times 2x}{(x^2+1)^2}$

$$= \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$$

$\left(\frac{1}{2} \right)$

$$y'' = \frac{(x^2+1)^2 \times -2x - (1-x^2) \times 2(x^2+1)' \times 2x}{(x^2+1)^4}$$

$$= \frac{-2x(x^2+1)^2 - 4x(1-x^2)(x^2+1)}{(x^2+1)^4}$$

$$= \frac{-2x(x^2+1) - 4x(1-x^2)}{(x^2+1)^3}$$

$$= \frac{-2x^3 - 2x - 4x + 4x^3}{(x^2+1)^3}$$

$$= \frac{2x^3 - 6x}{(x^2+1)^3}$$

$$= \frac{2x(x^2-3)}{(x^2+1)^3} \quad \left(\frac{1}{2} \right)$$

(b) (ii) Stat pts exist when $y' = 0$

Since $(x^2+1)^2 \neq 0$, $1-x^2 = 0$
 $x^2 = 1$
 $x = \pm 1$

When $x = +1$, $y = \frac{1}{2}$ $(1, \frac{1}{2})$

When $x = -1$, $y = -\frac{1}{2}$ $(-1, -\frac{1}{2})$

At $(1, \frac{1}{2})$ $y'' = \frac{2(1-3)}{84} = -\frac{2}{4} < 0$ MAX. stat. pt. ①

At $(-1, -\frac{1}{2})$ $y'' = \frac{-2(1-3)}{84} = \frac{2}{4} > 0$ min stat. pt. ①

Inflections occur when $y'' = 0$ and $\exists$ a sign change
 $2x(x^2-3) = 0$ because $(x^2+1)^3 \neq 0$

$x = 0, x^2 - 3 = 0$
 $x^2 = 3$
 $x = \pm \sqrt{3}$

At $x = 0$, $y = \frac{0}{0+1} = 0 \Rightarrow (0, 0)$ ①

At $x = +\sqrt{3}$, $y = \frac{\sqrt{3}}{4} \Rightarrow (\sqrt{3}, \frac{\sqrt{3}}{4})$ ①

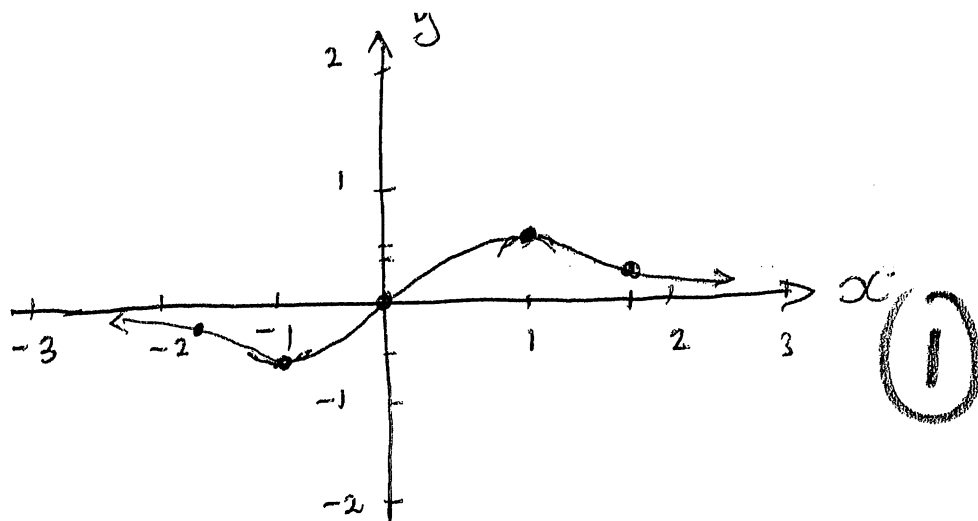
At $x = -\sqrt{3}$, $y = -\frac{\sqrt{3}}{4} \Rightarrow (-\sqrt{3}, -\frac{\sqrt{3}}{4})$ ①

At $x = 0 - \epsilon$ $y'' > 0$
 $x = 0 + \epsilon$ $y'' < 0$ } sign change

At $x = \sqrt{3} - \epsilon$ $y'' < 0$
 $x = \sqrt{3} + \epsilon$ $y'' > 0$ } sign change

At $x = -\sqrt{3} - \epsilon$ $y'' < 0$
 $x = -\sqrt{3} + \epsilon$ $y'' > 0$ } sign change

b) (iii)



$(1, \frac{1}{2})$ MAX

$(-1, -\frac{1}{2})$ MIN

$(0, 0)$ infL.

$(\sqrt{3}, \frac{\sqrt{3}}{4}) \doteq (1.7, .4)$ infL

$(-\sqrt{3}, -\frac{\sqrt{3}}{4}) \doteq (-1.7, -.4)$ infL

(c) $\frac{4}{5-x} \geq 1$ now $x \neq 5$.

$$\frac{4(5-x)^2}{(5-x)^2} \geq 1 \cdot (5-x)^2$$

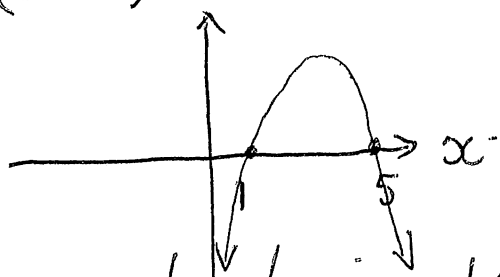
$$4(5-x) \geq (5-x)^2$$

$$4(5-x) - (5-x)^2 \geq 0$$

$$(5-x)[4 - (5-x)] \geq 0$$

$$(5-x)[-1+x] \geq 0$$

So $(x-1)(5-x) \geq 0$



above or touching $1 \leq x \leq 5$

but $x \neq 5$, so $\{x: 1 \leq x < 5\}$ //

3

2 (d) in radians

(4)

$$\text{If } \cos \theta = \cos \alpha$$

$$\theta = 2\pi \times n \pm \alpha$$

$$\text{So } \cos(\theta + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$$

$$\theta + \frac{\pi}{6} = 2\pi \times n \pm \frac{\pi}{6}$$

$$\theta = 2\pi n \pm \frac{\pi}{6} - \frac{\pi}{6}$$

$$\text{So } \theta = 2\pi n, \theta = 2\pi n - \frac{\pi}{3}$$

(4)

$$2(e) \text{ (i) } \sin \theta + \sqrt{3} \cos \theta = 2 \left(\frac{1}{2} \sin \theta + \frac{\sqrt{3}}{2} \cos \theta \right) \quad r = \sqrt{1^2 + \sqrt{3}^2} = 2.$$

$$\text{no } 2 \left(\frac{1}{2} \sin \theta + \frac{\sqrt{3}}{2} \cos \theta \right)$$

$$= R (\sin \theta \cos \alpha + \cos \theta \sin \alpha)$$

$$\text{now } R=2, \quad \left. \begin{array}{l} \cos \alpha = \frac{1}{2} \text{ quad } 1, 4 \\ \sin \alpha = \frac{\sqrt{3}}{2} \text{ quad } 1, 2 \end{array} \right\} \text{quad } 1.$$

$$\text{So } \alpha = \frac{\pi}{3} \text{ (radian measure)}$$

$$\text{So } 2 \sin(\theta + \frac{\pi}{3}) = \sin \theta + \sqrt{3} \cos \theta$$

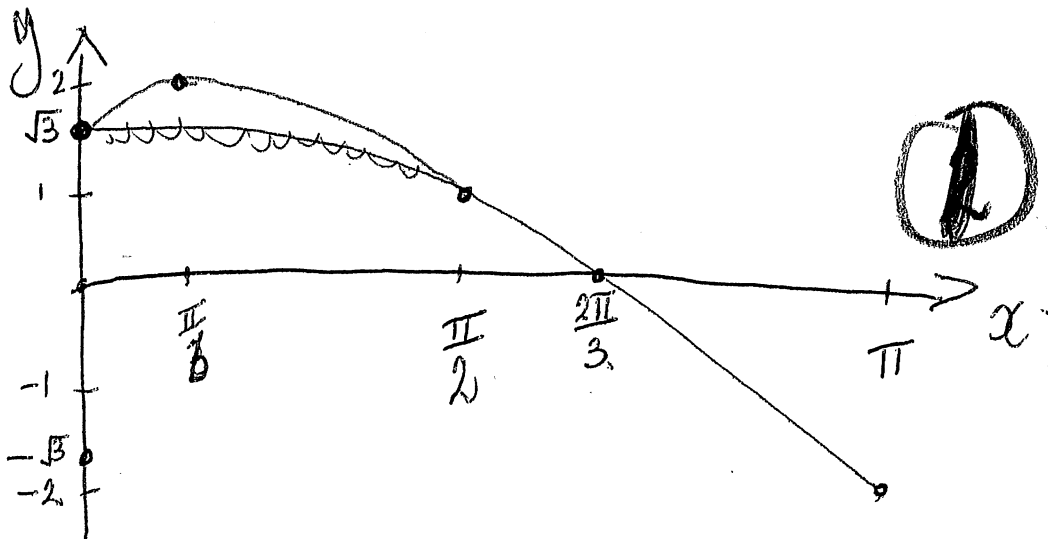
(ii)

$$y = 2 \sin(\theta + \frac{\pi}{3})$$

$$\theta = 0, y = 2 \sin \frac{\pi}{3} = \sqrt{3}$$

$$\theta = \frac{\pi}{2}, y = 2 \sin \frac{5\pi}{6} = 1$$

$$\theta = \pi, y = 2 \sin \frac{4\pi}{3} = -\sqrt{3}$$



Question [3] Solution.

$$\frac{\cos(\alpha+\beta)}{\cos(\alpha-\beta)} = \frac{4}{5}$$

(8)

$$5[\cos\alpha\cos\beta - \sin\alpha\sin\beta]$$

$$= 4[\cos\alpha\cos\beta + \sin\alpha\sin\beta]$$

$$\cos\alpha\cos\beta = 9\sin\alpha\sin\beta. \quad [2]$$

$$\therefore \tan\alpha\tan\beta = \frac{1}{9} \quad \text{--- (1)}$$

$$\text{If } PR = QR \Rightarrow \alpha + \beta = 45^\circ$$

$$\therefore \tan(\alpha + \beta) = 1 \quad [1]$$

$$\text{Now, } \tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = 1$$

$$\therefore \tan\alpha + \tan\beta = 1 - \tan\alpha\tan\beta$$

$$= 1 - \frac{1}{9}$$

$$= 8/9. \quad [2]$$

$$\Rightarrow 9(\tan\alpha + \tan\beta) = 8 \quad \text{--- (2)}$$

Let $\tan\alpha, \tan\beta$ be the roots to $x^2 - \frac{8}{9}x + \frac{1}{9} = 0$ [1]

$$\therefore 9x^2 - 8x + 1 = 0$$

$$x = \frac{8 \pm \sqrt{64-36}}{18}$$

[2]

$$\therefore \tan\alpha = \frac{8 + 2\sqrt{7}}{18} \quad \alpha = 36^\circ 27'$$

$$\text{or } \tan\beta = \frac{8 - 2\sqrt{7}}{18} \quad \beta = 8^\circ 33'$$

(6)

$$p(x) = (x-9)^2 q(x).$$

(4)

$$(i) \quad p'(x) = 2q(x)(x-9) + (x-9)^2 q'(x)$$

$$\therefore p'(9) = 0 \quad [2]$$

$$(ii) \quad p(1) = 0 \Rightarrow 1 + a + b - 4 = 0$$

$$\therefore a + b = 3 \quad \text{--- (1)} \quad [2]$$

$$p'(1) = 0 \quad 4 + 3a + 2b - 5 = 0$$

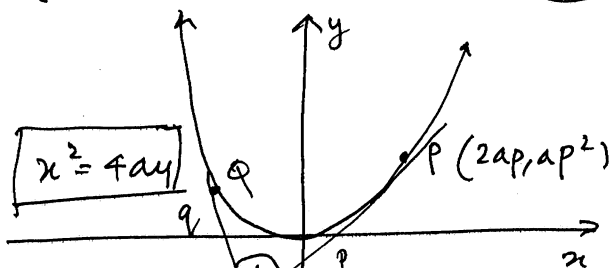
$$\text{i.e. } 3a + 2b = 5 \quad \text{--- (2)}$$

$$\textcircled{1} \times [-2] \quad -2a + 2b = -6 \quad \text{--- (3)}$$

$a = -1$
$b = 4$

(c)

(8)



$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

$$\left. \frac{dy}{dx} \right|_{x=2ap} = \frac{2ap}{2a} = p$$

$$(i) \quad y - ap^2 = p(x - 2ap) \quad [1]$$

$$y = px - ap^2 \quad \text{--- (1)}$$

Equation of QT is

$$(ii) \quad y = qx - aq^2 \quad \text{--- (2)}$$

Solve (1) & (2)

$$(p-q)x = a(p+q)(p-q) \quad [2]$$

$$\therefore \left. \begin{aligned} x &= a(p+q) \\ y &= apq \end{aligned} \right\} \quad \text{--- (4)}$$

$$\tan\alpha = 1$$

$$(\tan\alpha = \frac{p-q}{1+pq}) \quad [2]$$

$$\therefore p-q = 1+pq \quad \text{--- (5)}$$

$$\frac{x}{a} = p+q$$

$$\frac{y}{a} = pq \quad [3]$$

$$(p-q)^2 = (p+q)^2 - 4pq$$

$$\therefore (1+pq)^2 = (p+q)^2 - 4pq$$

$$\therefore \left(1 + \frac{y}{a}\right)^2 = \frac{x^2}{a^2} - \frac{4y}{a}$$

$$1 + \frac{2y}{a} + \frac{y^2}{a^2} = \frac{x^2}{a^2} - \frac{4y}{a}$$

$$a^2 + 2ay + y^2 = x^2 - 4ay \quad \text{--- (6)}$$

Locus of T

$$x^2 - y^2 - 6ay - a^2 = 0$$